

Chapter 57: Transforming Higher Education through Artificial Intelligence: A Study of Students' Perceptions and Learning Experiences

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Abstract

Artificial Intelligence (AI) is rapidly changing the way teaching and learning take place in higher education. AI-powered tools such as ChatGPT, Google Gemini, Microsoft Copilot, Grammarly, and Canva AI have become valuable learning companions, helping students complete academic tasks more efficiently and improve their overall learning experience. As these technologies become more accessible, it is important to understand how students use them and what they think about their role in education. This chapter explores students' perceptions, usage patterns, and expectations regarding AI-powered tools through a survey of 500 undergraduate, postgraduate, and research students from different academic disciplines. The findings show that most students regularly use AI for assignment preparation, project work, examination preparation, research, and understanding difficult concepts. Students believe that AI enhances personalized learning, provides quick access to information, offers instant feedback, saves time, and improves academic productivity. The study also reveals strong support for introducing AI literacy, prompt engineering, ethical AI, and digital skills into higher education curricula. Most respondents believe that AI will become an essential part of future teaching and learning. The chapter concludes that AI has significant potential to enrich higher education when used responsibly. It emphasizes the need for educational institutions to promote AI literacy, establish ethical guidelines, and integrate AI into teaching practices in ways that support learning while maintaining academic integrity and critical thinking.

Keywords: Artificial Intelligence, Higher Education, AI-Powered Tools, Student Perception, Digital Learning, Educational Technology.

1. Introduction

Artificial Intelligence (AI) has become one of the most significant technological developments in recent years, bringing remarkable changes to the field of higher education. With the emergence of generative AI tools, students now have access to personalized learning, instant academic support, automated feedback, intelligent tutoring, and a wide range of interactive learning resources. Unlike traditional digital learning platforms, AI-powered applications can generate content, explain difficult concepts, assist with writing, analyze data, and provide real-time support, making learning more engaging and accessible. Higher education institutions are increasingly embracing AI to enhance teaching and learning practices. AI-powered tools such as ChatGPT, Google Gemini, Microsoft Copilot, Grammarly, Canva AI, and AI-enabled learning management systems are now widely used by students for a variety of academic purposes. These tools help learners prepare assignments and presentations, conduct literature reviews, solve numerical and programming problems, improve writing skills, analyze data, and prepare for examinations. By reducing the time required for routine academic tasks, AI enables students to focus more on understanding concepts, critical thinking, and creative problem-solving. While AI offers numerous educational benefits, its growing use also raises important concerns. Issues such as academic integrity, plagiarism, overdependence on AI-generated content, misinformation, algorithmic bias, and data privacy have become major topics of discussion among educators and policymakers. As a result, universities around the world are introducing AI literacy programmes, ethical guidelines, and institutional policies to promote the responsible and effective use of AI in education. Understanding how students perceive and use AI-powered tools is essential for designing effective teaching strategies and preparing graduates for an increasingly digital world. Against this backdrop, this

chapter examines students' awareness, usage patterns, perceived benefits, and future expectations of AI-powered tools in higher education. The findings provide valuable insights that can help educational institutions integrate AI into teaching and learning in a way that enhances student learning while encouraging ethical, responsible, and informed use of these technologies.

2. Objectives of the Study

1. To examine the level of awareness and adoption of AI-powered tools among higher education students.
2. To identify the most frequently used AI applications for academic purposes.
3. To analyze the frequency and purpose of AI usage in learning activities.
4. To examine students' perceptions regarding the educational benefits of AI-powered tools.
5. To assess students' opinions regarding AI literacy and AI-related curriculum integration.
6. To examine students' expectations regarding the future role of AI in higher education.

3. Review of Literature

Kasneci et al. (2023) highlighted that generative AI tools such as ChatGPT have the potential to transform education by supporting personalized learning, adaptive tutoring, automated assessment, and academic writing. However, the authors also emphasized the importance of ethical guidelines, transparency, and responsible AI use in educational settings.

Lo (2023) examined the educational applications of ChatGPT and reported that AI significantly improves students' learning efficiency, creativity, and writing support. The study recommended that higher education institutions integrate AI literacy into teaching and learning while addressing concerns related to plagiarism and academic honesty.

Dwivedi et al. (2023) discussed the broader implications of generative AI for business, education, and society. Their research suggested that AI enhances decision-making, knowledge creation, and productivity but requires appropriate governance frameworks to manage ethical, legal, and social challenges.

UNESCO (2023) published global guidance on the use of generative AI in education and research, emphasizing the need for human-centered AI, digital inclusion, data privacy, transparency, and responsible implementation. UNESCO recommended that educational institutions develop AI policies and promote AI literacy among students and educators.

The **World Economic Forum (2025)** identified AI literacy, digital literacy, analytical thinking, creativity, and technological competence among the most important skills required for the future workforce. The report highlighted the growing importance of integrating AI-related competencies into higher education curricula to enhance graduate employability.

Data Analysis Table 1. Demographic Profile of Respondents(Source: Primary Data)

The table show participants were male (52.0%), while 47.0% were female. Most respondents (63.0%) belonged to the 18–21 years age group. Undergraduate students (66.0%) formed the largest proportion of the sample. Among the disciplines, Commerce (30.0%) and Science (28.0%) had the highest representation. In terms of institution type, 36.0% of the respondents were from government institutions, followed by 34.0% from private institutions and 28.0% from autonomous institutions. Regarding location, 46.0% of the respondents studied in urban institutions, 34.0% in semi-urban institutions, and 20.0% in rural institutions. Overall, the sample represents students from diverse educational backgrounds and institutional settings, making it suitable for the study.

Table 2 AI Usage and Awareness among Respondents

Variable	Category	Frequency (N)	Percentage (%)
Variable	Category	Frequency (N)	Percentage (%)
Gender	Male	260	52.0
	Female	235	47.0
	Other	5	1.0
Age Group	Below 18 years	20	4.0
	18–21 years	315	63.0
	22–25 years	145	29.0
	Above 25 years	20	4.0
Level of Education	Undergraduate (UG)	330	66.0
	Postgraduate (PG)	130	26.0
	Research Scholar	30	6.0
	Other	10	2.0
Discipline / Stream	Commerce	150	30.0
	Science	140	28.0
	Arts/Humanities	70	14.0
	Management	60	12.0
	Engineering/Technology	70	14.0
	Other	10	2.0
Type of Institution	Government College/University	180	36.0
	Private College/University	170	34.0
	Autonomous Institution	140	28.0
	Other	10	2.0
Location of Institution	Urban	230	46.0
	Semi-Urban	170	34.0
	Rural	100	20.0
Have you used AI-powered tools for academic purposes?	Yes	455	91.0
	No	45	9.0
AI tools	ChatGPT	385	77.0

commonly used*	Google Gemini	245	49.0
	Microsoft Copilot	120	24.0
	AI Chatbots	215	43.0
	Grammarly	180	36.0
	Canva AI	205	41.0
	AI-based Learning Management Systems	95	19.0
	Virtual Tutors	85	17.0
	Other	15	3.0
Frequency of AI tool usage	Daily	165	33.0
	Several times a week	175	35.0
	Occasionally	110	22.0
	Rarely	35	7.0
	Never	15	3.0
Academic activities using AI	Assignment preparation	360	72.0
	Project work	290	58.0
	Examination preparation	265	53.0
	Research paper writing	175	35.0
	Presentation preparation	240	48.0
	Learning difficult concepts	320	64.0
	Data analysis	145	29.0
	Other	20	4.0
Source of awareness about AI tools	College/University training	95	19.0
	Friends and classmates	135	27.0
	Social media	105	21.0
	Online courses/tutorials	55	11.0
	Self-learning	100	20.0
	Other	10	2.0

(Source: Primary Data)

The table indicate that 91% of students have used AI-powered tools for academic purposes, with ChatGPT (77%) emerging as the most widely used AI tool, followed by Google Gemini (49%) and AI Chatbots (43%). Most students reported using AI daily or several times a week (68%), demonstrating regular integration of AI into their learning. AI was primarily used for assignment preparation (72%), learning difficult concepts (64%), and project work (58%). The main sources of awareness about AI tools were friends and classmates (27%), social media (21%), and self-learning (20%), indicating that peer networks and digital platforms play a significant role in promoting AI adoption among students.

Table 3 Students' Perception of the Benefits of AI in Learning

Statement	SD (1)	D (2)	N (3)	A (4)	SA (5)	Me an
AI tools help me understand AI tools help me difficult	15 (3.0 %)	30 (6.0 %)	65 (13.0 %)	220 (44.0 %)	170 (34.0 %)	4.0 0

concepts better.						
AI improves my academic productivity.	10 (2.0 %)	25 (5.0 %)	55 (11.0 %)	235 (47.0 %)	175 (35.0 %)	4.1 5
AI helps me complete assignments more efficiently.	12 (2.4 %)	28 (5.6 %)	60 (12.0 %)	225 (45.0 %)	175 (35.0 %)	4.0 5
AI improves my problem-solving ability.	20 (4.0 %)	35 (7.0 %)	85 (17.0 %)	215 (43.0 %)	145 (29.0 %)	3.8 6
AI supports personalized learning according to my needs.	18 (3.6 %)	30 (6.0 %)	72 (14.4 %)	225 (45.0 %)	155 (31.0 %)	3.9 4
AI provides faster access to learning resources.	8 (1.6 %)	18 (3.6 %)	49 (9.8%)	235 (47.0 %)	190 (38.0 %)	4.1 6
AI increases my motivation and interest in learning.	18 (3.6 %)	42 (8.4 %)	90 (18.0 %)	210 (42.0 %)	140 (28.0 %)	3.8 2

(Source: Primary Data)

The above table indicate that students generally have a positive perception of AI in education. Most respondents agreed or strongly agreed that AI helps them understand difficult concepts, improves academic productivity, enhances assignment completion, and provides quicker access to learning resources. The highest mean score was recorded for "AI provides faster access to learning resources" (Mean = 4.16), followed by "AI improves my

academic productivity" (Mean = 4.15). Overall, the mean scores ranging from 3.82 to 4.16 indicate that students perceive AI as an effective tool for enhancing learning, productivity, and academic performance

Table 4 Student Perception Regarding the Benefits of AI in Education

Variable	Category	Frequency (N)	Percentage (%)
Major benefits of AI in education	Personalized learning	320	64.0
	Instant feedback	285	57.0
	24×7 learning support	305	61.0
	Better access to information	375	75.0
	Time saving	340	68.0
	Improved creativity	245	49.0
	Better exam preparation	275	55.0
	Career skill development	260	52.0
AI-powered tools have improved my overall learning experience.	Strongly Disagree	12	2.4
	Disagree	28	5.6
	Neutral	60	12.0
	Agree	225	45.0
	Strongly Agree	175	35.0
	Mean Score	4.05	
AI helps students learn at their own pace.	Strongly Disagree	10	2.0
	Disagree	25	5.0
	Neutral	65	13.0
	Agree	230	46.0
	Strongly Agree	170	34.0
	Mean Score	4.05	

(Source: Primary Data)

The above table show that most frequently reported benefits were better access to information (75.0%), time saving (68.0%), personalized learning (64.0%), and 24×7 learning support (61.0%). More than half of the respondents also recognized AI's role in instant feedback, better exam preparation, and career skill development. Furthermore, the majority of students agreed or strongly agreed that AI-powered tools have improved their overall learning experience and enabled them to learn at their own pace, with both statements recording a mean score of 4.05. These findings indicate a strong positive perception of the educational benefits of AI among students.

Table 5 Students' Opinion on AI Literacy and Future of AI in Higher Education

Variable	Category	Frequency (N)	Percentage (%)
Should colleges/universities provide AI literacy training to students?	Yes	445	89.0
	No	20	4.0
	Not Sure	35	7.0
AI-related skills to be included in the curriculum*	AI Literacy	390	78.0
	Prompt	315	63.0

	Engineering		
	Data Analytics	290	58.0
	Ethical Use of AI	350	70.0
	Digital Skills	330	66.0
	AI Applications in Subject Areas	340	68.0
How can institutions encourage responsible AI usage?*	Conduct AI awareness programs	335	67.0
	Provide AI training workshops	365	73.0
	Develop AI usage guidelines	285	57.0
	Include AI courses in curriculum	345	69.0
	Promote ethical AI practices	355	71.0
AI will become an essential part of higher education in the future.	Strongly Disagree	8	1.6
	Disagree	17	3.4
	Neutral	55	11.0
	Agree	225	45.0
	Strongly Agree	195	39.0
	Mean Score	4.16	

(Source: Primary Data)

4. Major Findings

1. The study found that 91% of the respondents had used AI-powered tools for academic purposes, indicating widespread adoption of AI in higher education.
2. ChatGPT (77%) emerged as the most widely used AI tool, followed by Google Gemini (49%), AI Chatbots (43%), Canva AI (41%), and Grammarly (36%).
3. Most students (68%) reported using AI tools either daily or several times a week, demonstrating that AI has become an integral part of their academic routine.
4. Students primarily used AI for assignment preparation (72%), learning difficult concepts (64%), project work (58%), examination preparation (53%), and presentation preparation (48%).
5. The major sources of awareness about AI tools were friends and classmates, social media, self-learning, and institutional training programmes.
6. Students expressed a positive perception of AI-powered tools. The highest-rated benefits were faster access to learning resources, improved academic productivity, efficient assignment completion, and enhanced understanding of difficult concepts.
7. The most frequently reported educational benefits of AI included better access to information, time saving, personalized learning, 24×7 learning support, instant feedback, and better examination preparation.
8. A large majority (89%) believed that colleges and universities should introduce AI literacy training as part of higher education.
9. Students recommended including AI Literacy, Prompt Engineering, Ethical AI, Digital Skills, Data Analytics, and subject-specific AI applications in the curriculum.
10. Most respondents believed that Artificial Intelligence will become an essential component of future higher education and significantly influence teaching, learning, assessment, and research.

5. Suggestions

1. Higher education institutions should introduce AI literacy programmes to develop students' understanding of AI technologies and their responsible use.

2. Universities should integrate AI-related topics such as Prompt Engineering, Ethical AI, Data Analytics, Digital Skills, and discipline-specific AI applications into undergraduate and postgraduate curricula.
3. Faculty members should receive regular professional development and training to effectively integrate AI-powered tools into teaching and assessment.
4. Institutions should formulate clear policies and ethical guidelines regarding the use of AI in academic work to maintain academic integrity and reduce plagiarism.
5. Students should be encouraged to use AI as a learning support tool rather than a substitute for independent thinking, creativity, and problem-solving.
6. Workshops, seminars, and awareness programmes should be organized to promote responsible, ethical, and informed use of AI technologies.
7. Educational institutions should invest in AI-enabled digital infrastructure and learning management systems to provide equitable access to AI resources for all students.
8. Future research should investigate the long-term impact of AI on learning outcomes, critical thinking, academic performance, and employability across different disciplines and educational settings.

7. Conclusion

Artificial Intelligence (AI) is becoming an integral part of higher education and is transforming the way students learn, conduct research, and complete academic tasks. The findings of this study show that students widely use AI-powered tools and consider them helpful for improving learning, saving time, understanding complex concepts, and increasing academic productivity. Among the various AI applications, ChatGPT was the most commonly used tool, highlighting the growing importance of generative AI in students' daily academic activities. The study also found that students strongly support the inclusion of AI literacy and ethical AI practices in higher education curricula. They believe that learning how to use AI responsibly is an essential skill for their future academic and professional careers. Although AI offers many benefits, students also recognize the importance of using these technologies responsibly and ethically. Educational institutions should establish clear policies and provide proper guidance to ensure that AI is used to support learning without compromising academic integrity, critical thinking, or creativity. Overall, AI should be viewed as a tool that complements the teaching and learning process rather than replacing teachers or students' independent efforts. By encouraging responsible AI use and strengthening AI literacy, higher education institutions can equip students with the knowledge, skills, and ethical awareness needed to succeed in an increasingly digital and AI-driven world.

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